GRC & OPERATIONAL RISK PORTFOLIO

GRAFANA IMPLEMENTATION COMPLETION REPORT

Enterprise GRC Risk Monitoring Dashboard
Eight completed monitoring panels

Prepared for	Phani Kumar
Implementation status	**Completed**
Environment	Grafana on Docker Desktop (Windows)
Report date	04 November 2026

Project purpose

Document the completed Grafana deployment and the eight-panel GRC risk monitoring dashboard developed as part of the small-scale manufacturing risk management portfolio.

1. Executive Summary

Grafana was implemented as the visualization layer for the project's Enterprise Risk Management (ERM), Operational Risk Management (ORM), Risk and Control Self-Assessment (RCSA), and SaaS risk/KRI analysis. The completed deliverable is **one Enterprise GRC Risk Monitoring Dashboard containing eight completed visualization panels**.

The dashboard brings together headline operational risk volume, risk severity, risk categories, the highest exposures, control effectiveness, residual risk, treatment progress and operational KRI monitoring. The Grafana environment was deployed locally using Docker Desktop, with persistent storage configured through a Docker volume. The project includes a portfolio screenshot in the GitHub documentation folder.

2. Objectives and Scope

• Create a practical visual monitoring layer for the ERM/ORM and RCSA work completed for the manufacturing case study.

• Make risk severity, categories, exposure and treatment status easier to review.

• Show control effectiveness and residual risk in visual form.

• Monitor operational KRIs such as machine downtime, production delays and supplier exceptions.

• Produce a demonstrable Grafana implementation for the GRC portfolio.

3. Completed Dashboard — Eight Panels

The following eight panels form the completed **Enterprise GRC Risk Monitoring Dashboard**. These are eight panels within the dashboard, not eight separate Grafana dashboards.

No.	Panel name	Visualization	Monitoring purpose
1	**Total Operational Risks**	Stat	Headline count of identified operational risks (18 risks in the project dataset).
2	**Risk Severity Distribution**	Pie chart	Distribution across Critical, High, Medium and Low severity levels.
3	**Operational Risk Categories**	Bar chart	Risk distribution across production, equipment, supply chain, financial and fraud categories.
4	**Top 5 Risk Exposures**	Table	Prioritised view of the five highest risk exposures.
5	**Control Effectiveness**	Gauge	Visual summary of control effectiveness.
6	**Residual Risk Assessment**	Donut chart	Distribution of residual risk after considering controls.
7	**Risk Treatment Status**	Bar chart	Treatment progress across Accept, Mitigate and Transfer decisions.
8	**Operational KRI Monitoring**	Time series	Trend monitoring for machine downtime, production delays, supplier exceptions and other operational KRIs.

4. Technical Environment

Component	Implementation detail
Platform	Grafana
Deployment	Docker Desktop on Windows
Container image	grafana/grafana
Local interface	http://localhost:3000

Component	Implementation detail
Persistent storage	Docker volume: grafana-storage
Data source	GRC Risk Data (Infinity)
Input approach	Prepared CSV risk and KRI datasets

5. Data Inputs and Monitoring Logic

The dashboard uses prepared data extracts derived from the project's risk registers, RCSA records and SaaS/operational KRI work. The Grafana data source was configured for the prepared risk data. The implementation should be understood as a file-based analytical dashboard, rather than a live transactional integration with Eramba or manufacturing systems.

• ERM and operational risk register data.

• Risk and Control Self-Assessment (RCSA) data.

• SaaS risk register and SaaS RCSA data where relevant to the portfolio view.

• Operational KRI history and status data, including downtime, production delays and supplier exceptions.

6. Deployment and Configuration Activities

• Pulled and deployed the Grafana container using the grafana/grafana image.

• Exposed the Grafana web interface through local port 3000.

• Configured the grafana-storage Docker volume to retain Grafana application data.

• Accessed Grafana and changed the initial administrator password.

• Configured the GRC Risk Data (Infinity) data source for the prepared project datasets.

• Created the Enterprise GRC Risk Monitoring Dashboard and completed the eight monitoring panels listed in Section 3.

• Captured a dashboard screenshot and stored it in the GitHub portfolio repository.

7. Validation and Completion Evidence

The Grafana application was accessed successfully in the local Docker environment. Persistent storage was configured, the project data source was added, and the eight dashboard panels were completed as part of the portfolio implementation. The repository contains the screenshot at:

docs/03_Data_Visualization/03_Grafana/Grafana Portfolio.png

The screenshot is visual portfolio evidence of the Grafana work. The underlying project workbook/CSV inputs remain the source of the risk and KRI values presented in the dashboard.

8. Relationship to the Wider GRC Portfolio

Grafana complements the Excel-based ERM/RCSA and SaaS risk analysis and the separate Power BI and Tableau visualization work. Grafana is maintained as an independent visualization layer. Power BI and Tableau were not required to connect to Eramba, and this report does not claim a live Grafana-to-Eramba integration.

9. Scope Boundaries and Limitations

• This is a local Docker-based portfolio implementation, not a hardened production deployment.

• The dashboard uses prepared CSV-based analytical inputs; it is not a live feed from production machinery, ERP systems or Eramba.

• Automated alerting, external notifications and scheduled data refresh are not claimed as completed in this report.

• The dashboard provides risk and KRI visualization; it does not implement Basel III capital calculations or a formal stress-testing engine.

• Dashboard accuracy depends on the completeness, quality and refresh of the source datasets.

10. Handover and Operating Notes

• Start Docker Desktop before launching the local Grafana environment.

• Confirm the Grafana container is running and open http://localhost:3000.

• Use the project owner's securely stored administrator credentials.

• When risk registers or KRI history are updated, refresh the relevant prepared data files and validate the dashboard panels.

• Retain updated screenshots and, where available, an exported dashboard JSON as versioned implementation evidence.

11. Completion Statement

The Grafana component is recorded as **completed** for the defined portfolio scope: a Docker-hosted Grafana environment with persistent storage, a configured project risk-data source and one Enterprise GRC Risk Monitoring Dashboard containing eight completed panels. The implementation demonstrates practical experience in Grafana deployment, data-source setup, risk visualization and operational KRI trend monitoring. This completion statement is limited to the scope and evidence described in this report.

12. Repository Reference

GitHub repository: phaniirukuvarjula-lgtm/phaniirukuvarjula-lgtm.github.io
Folder: docs/03_Data_Visualization/03_Grafana/
Screenshot: Grafana Portfolio.png
Final report: Grafana_GRC_KRI_Monitoring_Implementation_Completion_Report.pdf